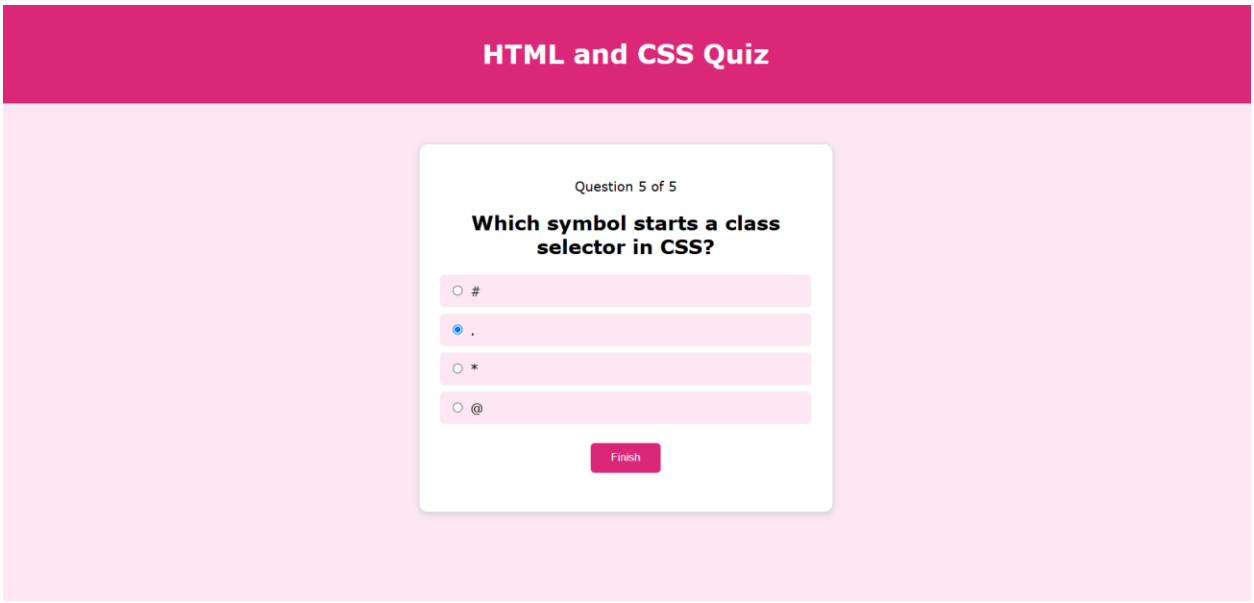
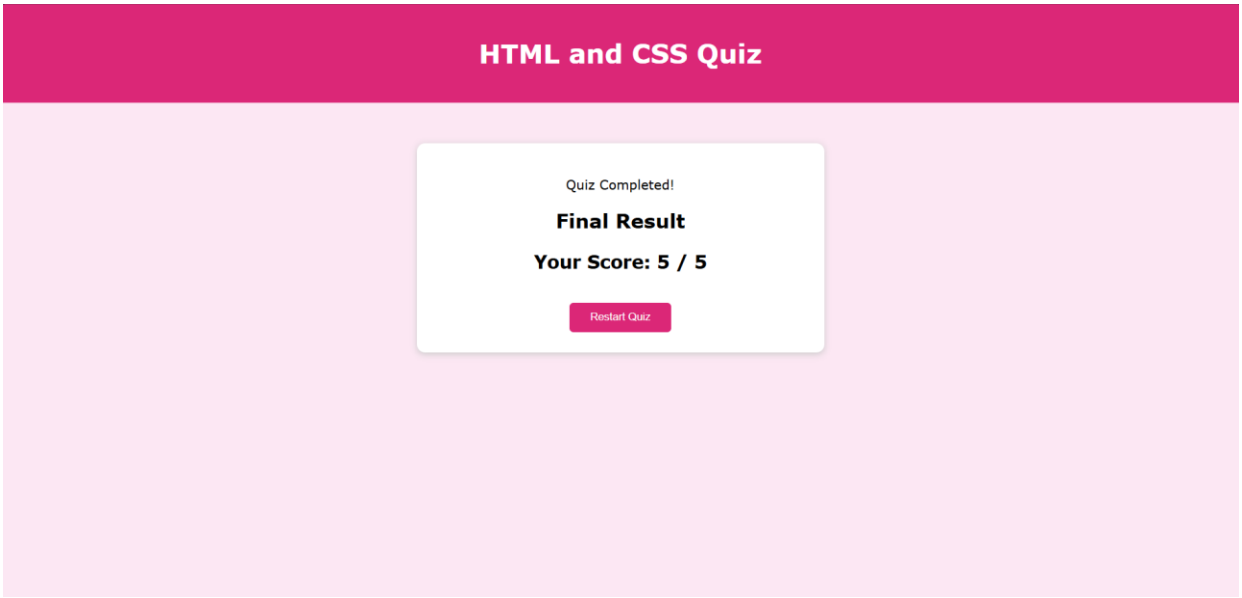
Lab 14 : Create a quiz application that displays one question at a time. Track the user's score and show the final result at the end.

TAGS / CSS USED	ATTRIBUTES / VALUES USED	DESCRIPTION
<!DOCTYPE html>	html	Defines an HTML5 document.
<html>	lang="en"	Root element of the webpage.
<head>	—	Contains title, CSS, and metadata.
<meta>	charset, viewport	Defines character encoding and responsive display.
<title>	Simple Quiz Application	Sets the browser tab title.
<style>	CSS rules	Adds internal CSS styling.
<body>	—	Contains the visible webpage content.
<header>	—	Creates the quiz header.
<h1>, <h2>	id="question"	Displays the quiz title and questions.
	class="quiz-box", id="options"	Creates the quiz container and options section.
<p>	id="number", id="result"	Displays question number and final score.
<button>	onclick, id	Creates Next and Restart buttons.
<label>	class="option"	Displays each answer option.
<input>	type="radio", name="answer"	Allows users to select one answer.
	id="option0" etc.	Displays the answer option text.
<script>	JavaScript code	Contains quiz logic and functions.
.quiz-box	max-width, padding, margin	Styles and centers the quiz container.
.option	display: block, cursor: pointer	Styles clickable answer options.
border-radius	5px, 10px	Creates rounded corners.

box-shadow	0 2px 8px	Adds a shadow around the quiz box.
text-align	center, left	Aligns text inside elements.
button:hover	opacity: 0.85	Adds a hover effect to buttons.
var	questions, current, score	Declares JavaScript variables.
questions = []	Array of objects	Stores questions, options, and correct answers.
function showQuestion()	—	Displays the current question and options.
function nextQuestion()	—	Checks the answer and moves to the next question.
function restartQuiz()	—	Resets the quiz and score.
document.getElementById()	Element ID	Accesses HTML elements using their IDs.
document.querySelector()	input[name="answer"]:checked	Finds the selected radio button.
innerHTML, textContent	HTML / Text	Updates questions, options, and results dynamically.
for	i = 0; i < 4; i++	Loops through the four answer options.
if...else	Conditions	Checks answers and controls quiz flow.
Number()	selected.value	Converts the selected answer value into a number.
score++, current++	Increment operator	Increases the score or question number.
alert()	Message	Displays a popup when no answer is selected.
style.display	none, inline-block	Hides or displays buttons.
return	—	Stops execution when no answer is selected.

Final Outcome:





Code Snipet :

```
<!DOCTYPE html>
<html lang="en">

<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Simple Quiz Application</title>
 <style>
  body {
   font-family: Verdana;
   background: #fce7f3;
   text-align: center;
   margin: 0;
  }

  header {
   background: #db2777;
   color: white;
   padding: 20px;
  }

  .quiz-box {
   background: white;
   max-width: 450px;
   margin: 50px auto;
   padding: 25px;
   border-radius: 10px;
   box-shadow: 0 2px 8px #cccccc;
  }

  .option {
   display: block;
   text-align: left;
   padding: 10px;
   margin: 8px 0;
   background: #fce7f3;
   border-radius: 5px;
   cursor: pointer;
```

```
  }

  button {
    background: #db2777;
    color: white;
    border: none;
    padding: 10px 25px;
    margin-top: 15px;
    cursor: pointer;
    border-radius: 5px;
  }

  button:hover {
    opacity: 0.85;
  }

  #result {
    font-size: 22px;
    font-weight: bold;
  }
 </style>
</head>

<body>
 <header>
  <h1>HTML and CSS Quiz</h1>
 </header>
 <div class="quiz-box">
  <p id="number"></p>
  <h2 id="question"></h2>
  <div id="options"></div>
  <button id="nextBtn" onclick="nextQuestion()">Next</button>
  <p id="result"></p>
  <button id="restartBtn" onclick="restartQuiz()" style="display:none;">Restart
Quiz</button>
 </div>

 <script>
  var questions = [
   { question: 'What does HTML stand for?', options: ['Hyper Text Markup Language', 'High
Text Machine Language', 'Hyper Tool Markup Link', 'Home Text Main Language'], answer: 0
```

```
},
    { question: 'Which tag creates a paragraph?', options: ['<h1>', '<p>', '<br>', '<img>'], answer: 1 },
    { question: 'Which CSS property changes text color?', options: ['background', 'font-size', 'color', 'border'], answer: 2 },
    { question: 'Which tag adds an image?', options: ['<img>', '<a>', '<ul>', '<div>'], answer: 0 },
    { question: 'Which symbol starts a class selector in CSS?', options: ['#', '.', '*', '@'], answer: 1 }
  ];

  var current = 0;
  var score = 0;

  function showQuestion() {
    document.getElementById("number").innerHTML = "Question " + (current + 1) + " of " + questions.length;
    document.getElementById("question").textContent = questions[current].question;
    var html = "";
    for (var i = 0; i < 4; i++) {
      html += '<label class="option"><input type="radio" name="answer" value="' + i + '"> ';
      // text is added separately below for safe display of HTML tag options
      html += '<span id="option' + i + '"></span></label>';
    }
    document.getElementById("options").innerHTML = html;
    for (var i = 0; i < 4; i++) {
      document.getElementById("option" + i).textContent = questions[current].options[i];
    }
    document.getElementById("nextBtn").textContent = current === questions.length - 1 ? "Finish" : "Next";
  }

  function nextQuestion() {
    var selected = document.querySelector('input[name="answer"]:checked');
    if (selected == null) {
      alert("Please select an answer!");
      return;
    }
    if (Number(selected.value) === questions[current].answer) {
      score++;
    }
```

```
    current++;
    if (current < questions.length) {
      showQuestion();
    } else {
      document.getElementById("number").innerHTML = "Quiz Completed!";
      document.getElementById("question").innerHTML = "Final Result";
      document.getElementById("options").innerHTML = "";
      document.getElementById("nextBtn").style.display = "none";
      document.getElementById("result").innerHTML = "Your Score: " + score + " / " +
questions.length;
      document.getElementById("restartBtn").style.display = "inline-block";
    }
  }

  function restartQuiz() {
    current = 0;
    score = 0;
    document.getElementById("result").innerHTML = "";
    document.getElementById("nextBtn").style.display = "inline-block";
    document.getElementById("restartBtn").style.display = "none";
    showQuestion();
  }

  showQuestion();
 </script>
</body>

</html>
```